Paleomagnetic Results from the Mauch Chunk Formation: A Test of the Origin of Curvature in the Folded Appalachians of Pennsylvania[1]

RAYMOND R. KNOWLES AND NEIL D. OPDYKE

Lamont Geological Observatory, Columbia University, Palisades, New York 10964

Specimens of red siltstone were collected at twenty-five sites in the Upper Mississippian Mauch Chunk formation of Pennsylvania. The sites were so distributed as to test whether the curve in the folded Appalachians was formed after the magnetization of the beds or whether the orginal geosyncline in Mississippian time was curved in plan. All specimens were partially demagnetized to isolate the stable component. When the beds are rotated back into a horizontal position, the stable components show a decrease in the scatter between the sites, indicating that the direction of magnetization was acquired before the folding of the Appalachians. The directions of demagnetization were found to be the same around the curve of the Appalachians, apparently ruling out their horizontal rotation after the acquisition of their stable magnetization. The pole position calculated from the Mauch Chunk is at 43°N, 127°E with a $\Delta p = 3.1$, $\Delta m = 6.3$. This pole position is in general agreement with other late Paleozoic poles.

INTRODUCTION

Gwinn [1967] has listed several possible mechanisms for the origin of curvature in a folded mountain system. The end result of the several mechanisms can be visualized as a horizontal rotation. Whether such rotation has actually taken place in a mountain system can be determined from paleomagnetic studies [*Irving*, 1964].

The Appalachian salient in Pennsylvania was chosen to test, by means of paleomagnetic studies, the hypothesis that the curvature was produced by the bending of the folded mountain system. The Upper Mississippian Mauch Chunk red beds are suitable for paleomagnetic study and outcrop along the salient. Studies of this type have been attempted for the Silurian Bloomsburg formation [*Roy et al.*, 1967], but the results were inconclusive.

SAMPLING

The Mauch Chunk consists of dark red intercalated fine-grained sandstones, siltstones, and shales. It crops out (Figure 1) along the margin of the Broadtop syncline and throughout the Anthracite region at the southern and eastern ends, respectively, of the salient. The form of the salient is shown by the Allegheny structural front. Eleven sites were sampled in the Broadtop syncline, and fourteen sites were sampled in the Anthracite region. At each site three or four cores 2.34 cm in diameter and about 7.62 cm long were taken by means of a portable drill. The cores were oriented by means of a Brunton compass while still attached to the host rock, and then they were broken free. All sites were located in road cuts or in river beds.

LABORATORY PROCEDURE

In the laboratory the cores were cut into cylinders 2.34 cm long. The specimens at the bottom of the core were labeled A, and the next highest B, etc. To avoid the effects of weathering only A disks were used in this study. Each core provided one sample; therefore, each site contained three or four samples. The direction and intensity of remanent magnetization was measured on a 5-cps spinner magnetometer of the type developed by *Foster* [1966].

NATURAL REMANENT MAGNETIZATION

Figure 2 shows the NRM mean site directions corrected for bedding tilt for the twenty-five sites from the Mauch Chunk. North-pole-seeking vectors are indicated by crosses plotted on the lower hemisphere and by dots plotted on

[1] Lamont Geological Observatory Contribution 1239.

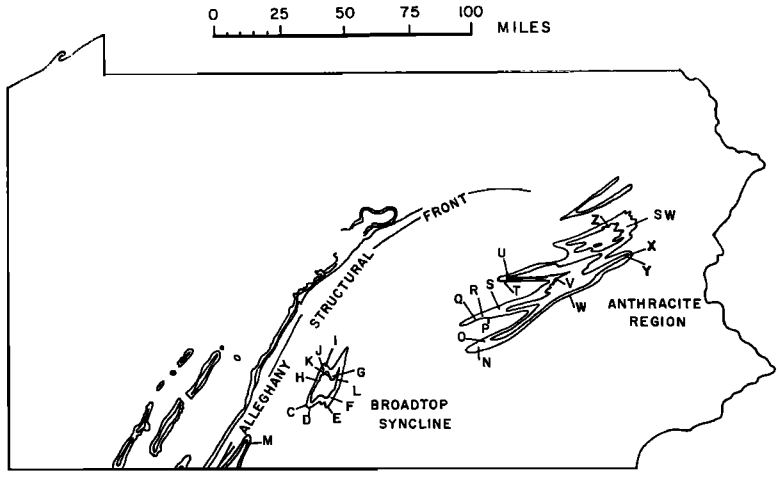


Fig. 1. Sketch map of Pennsylvania showing outcrop of Upper Mississippian Mauch Chunk red beds. Sampling sites indicated by a letter as in Table 1.

the upper hemisphere. Most vectors show a low inclination in a south-southeast direction. The NRM site mean intensities and directions of magnetization are given in Table 1.

Demagnetization

To determine whether the NRM direction is a stable direction or the resultant of two or more components of equal stability, step-wise thermal demagnetization [*Thellier*, 1938; *Irving and Opdyke*, 1965] was applied to all samples. The samples were placed in an oven and heated to a given temperature, held there for 15 to 20 minutes, and then allowed to cool in field-

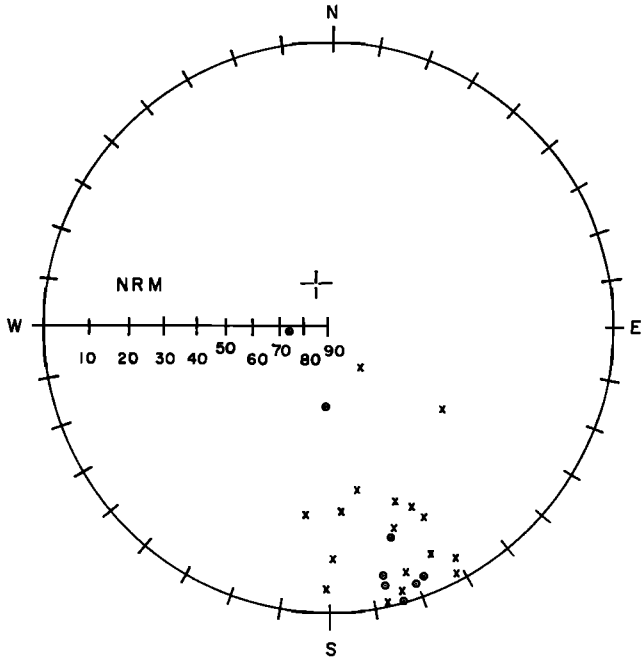


Fig. 2. NRM site mean directions corrected for bedding tilt plotted on lower hemisphere of Schmidt stereo net. North-seeking directions plotted, crosses (dots) plot on lower (upper) hemisphere. Present field position indicated by large broken cross.

21562202, 1968, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB073i020p06515 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Results from the Mauch Chunk Red Beds

	NRM						M_t					
Site	N	M	D_m, I_m	R	K	α_{95}	T_c, °C	M	D_m, I_m	R	K	α_{95}
MCC	4	0.248	153, +17	3.98	90.58	26.55	400	.256	159, +19	3.99	236.86	5.98
MCD	4	0.240	159, +24	3.90	31.29	16.68	550	.163	141, +17	3.82	16.63	23.21
MCE	4	0.348	159, −4	3.99	376.40	12.90	550	.064	162, +8	3.97	120.99	8.38
MCF	4	0.194	176, +24	3.89	28.92	17.37	550	.058	170, +13	3.95	64.61	11.51
MCG	4	0.112	151, +5	3.94	50.19	13.09	600	.016	164, −8	3.72	10.90	29.14
MCH	4	0.067	135, +38	3.77	13.20	26.26	600	.030	337, −9	3.94	50.10	13.10
MCI	4	0.097	142, +70	3.98	165.62	7.16	...	...	*	...	...	...
MCJ	4	0.087	170, +30	3.99	731.56	3.39	550	.035	162, +14	3.93	44.45	13.93
MCK	4	0.179	161, +17	3.95	65.13	11.46	550	.094	152, +11	3.96	68.44	11.18
MCL	4	0.071	180, +13	3.96	74.63	10.70	550	.018	188, +8	3.92	37.74	15.15
MCM	4	0.111	186, +23	3.75	12.05	27.59	...	...	*	...	...	...
					Mean direction Broadtop				162, +10	8.73	29.95	9.56
MCN	4	0.151	279, −73	3.92	37.85	15.12	...	...	*			
MCO	3	0.364	163, +4	2.98	179.48	6.43	550	.271	154, +27	2.99	270.68	7.50
MCP	4	0.198	154, +8	3.98	163.16	7.21	550	.075	155, −7	3.97	105.03	9.00
MCQ	4	0.879	155, +21	3.97	111.79	8.72	550	.234	160, +28	3.94	55.57	12.43
MCR	4	0.982	152, +2	3.94	47.46	13.47	550	.372	157, +10	3.98	162.64	7.22
MCS	4	0.495	163, −16	3.96	76.27	10.58	600	.052	155, −1	3.92	35.60	15.61
MCT	4	0.349	160, −4	3.99	704.60	3.46	...	...	*	...	...	...
MCU	4	1.000	167, +1	3.94	56.32	12.34	550	.270	158, +9	3.99	221.56	6.18
MCV	3	1.068	167, +10	2.99	1570.90	3.11	550	.828	167, +10	2.99	758.46	4.47
MCW	4	0.228	182, −69	3.16	3.61	56.59	...	...	*			
MCX	4	0.167	181, +5	3.91	35.34	15.67	550	.165	185, +1	3.90	30.46	16.91
MCY	4	0.167	162, +6	3.98	160.16	7.28	550	.151	161, +2	3.98	146.13	7.62
MCZ	4	0.669	164, −1	3.99	485.84	3.93	600	.591	167, −3	3.99	290.82	5.39
MCSW	4	0.249	167, −7	3.99	296.92	5.34	600	.134	171, −13	3.99	291.13	5.39
					Mean direction Anthracite region				163, +6	10.60	25.59	9.19
					Combined results				162, +8	19.32	28.21	6.25
					N = 20							

Notes

N is the number of oriented samples.
M is the mean intensity in emu $cm^{-3} \times 10^{-4}$.
D_m, I_m, is the site mean direction corrected for bedding tilt.
T_c is the cleaning temperature used.
R is the magnitude of the resultant.
K is Fisher's estimate of precision.
α_{95} is Fisher's circle of confidence (P = 0.05).
* Rejected site.

free space. The samples were measured, and the process was repeated at a higher temperature. During thermal demagnetization the geomagnetic field in the vicinity of the oven was constantly monitored and annulled.

The monitoring was performed by the following method. The oven occupied the center of a 3-coil square Helmholtz system. Three mutually perpendicular magnetometer probes were mounted near the oven, and they monitored the field. The magnetometers put out a signal proportional to the sensed field, and this signal was fed into three separate regulated dc power packs. The power supply has been programmed so that the output voltage follows the input signal and remains in a constant proportion to it. The output voltage was fed to the Hemholtz system. Transients and diurnal changes in the geomagnetic field were automatically cancelled. Mapping of the space that would be occupied by specimens during a run showed that the field intensity ranges from

21562202, 1968, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB073i020p06515 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

+160 to −130 gammas over a distance of about 20.32 cm.

Each sample was demagnetized in steps up to 550° or 650°C. The samples from the Broadtop syncline and the Anthracite region show no difference in behavior during demagnetization, and therefore no distinction according to site locality will be made in this part of the discussion. Five separate patterns of behavior can be recognized during thermal demagnetization. In the first group (nineteen sites) the declination of the stable component of magnetization differs by less than 15° from that of the NRM. However, the inclinations change, negative inclinations decreasing (Figure 3*A*) and positive inclinations increasing. In some cases (Figure 3*B*) the positive inclination becomes shallower at 300° and 400°C and then

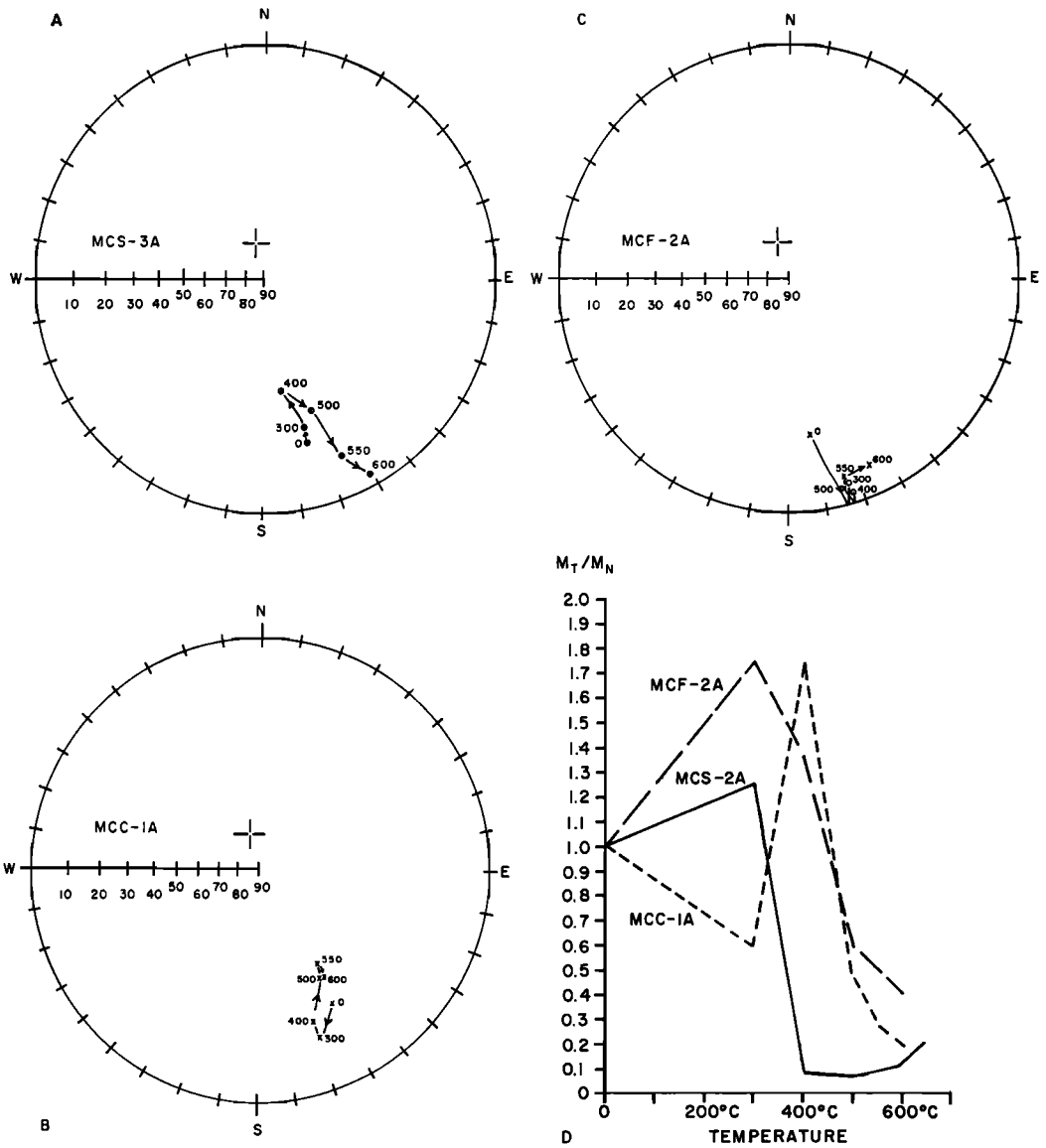


Fig. 3. Effect of thermal demagnetization on single samples from three separate sites showing the behavior pattern of group 1 data. Data corrected for bedding tilt. Present field position indicated by large broken cross.

21562202, 1968, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB073i020p06515 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

steepens again at 500°C. Several samples have a positive inclination for the NRM (Figure 3*C*), but the inclination becomes negative at 300° and 400°C and then becomes positive again at 500°C and remains so up to 600°C. All the thermal demagnetization curves show an initial increase of intensity, indicating that a vector opposed to the southeast direction is being removed. It seems reasonable to assume that the vector being removed at 300°C is a viscous remanence probably directed along the present geomagnetic field. Above 300°C a harder secondary component directed at a low angle to the horizontal and subparallel to the primary direction is being removed. A similar component has been recognized in Silurian rocks from the folded Appalachians by *Irving and Opdyke* [1965]. They believe it to be a Permian direction.

In the second group (one site), the direction vectors move along a great circle to a north-northwest position approximately 180° away from the data of group 1. Figure 4*A* shows the behavior of the sample from this group, and Figure 4*D* shows the thermal demagnetization curve. A secondary component directed along a south-southeast direction is being removed during the cleaning process, as is indicated by the low value for the intensity at 400°C and the increase in intensity when the vector moves into the northwest quadrant.

In the third group (three sites), the direction vectors show a western declination and a steep negative inclination. In sites MCM (Figure 4*B*) and MCT, the directions change along a partial great circle as step-wise thermal demagnetization is applied. In site MCN, the NRM shows a western declination and a steep negative inclination, and there is very little change in direction up to 625°C. The data from sites MCM, MCN, and MCT show an intermediate direction midway between the first two groups. This direction may be the resultant of two hard components of approximately equal intensity, having low negative inclinations and oppositely directed along a NNW–SSE line. On thermal demagnetization, specimens from sites MCM and MCT move toward the reversed direction but never reach a stable end point. This is analogous to the intermediate directions given by *Roy et al.* [1967]. *Roy and Robertson* [1968] have proposed a mechanism whereby steep intermediate directions in sedimentary rocks may be explained. The vector having an intermediate direction is the resultant of two approximately oppositely directed vectors. These two magnetization vectors are acquired in the following manner. A detrital remanent magnetization is acquired; subsequently, the field is reversed and during diagenesis a 'diagenetic' (chemical ?) remanent magnetization is acquired whose vector is approximately oppositely directed to the DRM vector. The resultant of these two vectors is an intermediate direction with a step inclination.

One site (MCI) shows a steep positive NRM and maintains this orientation up to 600°C (Figure 4*C*). Finally, there is one site that showed a lack of internal consistency during thermal demagnetization. The last two sites and the three sites from the third group are not included in the statistical analysis. The cleaning temperature chosen for the accepted sites was the temperature at which a stable direction of magnetization was reached. Often, but not always, the within-site dispersion was a minimum at this temperature.

Age of Magnetization

To determine the relative age of the stable paleomagnetic direction two variations of the fold test [*Graham*, 1949] were applied.

Test 1. Sites C and D were taken on the north and south limbs of a small anticline in the southwest corner of the Broadtop syncline. Figure 5*A* shows the direction vectors after demagnetization referred to the present horizontal. Figure 5*B* shows the same direction vectors corrected for bedding tilt. The two sites converge. Whether thermal demagnetization has removed substantial secondary components can be determined if reversals are present, since, if secondary components are added to bipolar vectors, the results will no longer be antiparallel [*Irving*, 1964]. In the upper left of Figure 5*B* are plotted the partially demagnetized data from site MCH, corrected for bedding tilt. The data are plotted on the upper hemisphere and are the reverse of the data of sites C and D. The reversal indicates that substantial secondary components are absent.

Test 2. Figure 6*A* is a plot of site mean directions of all thermally cleaned data referred to the horizontal. Figure 6*B* shows the same

21562202, 1968, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB073i020p06515 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

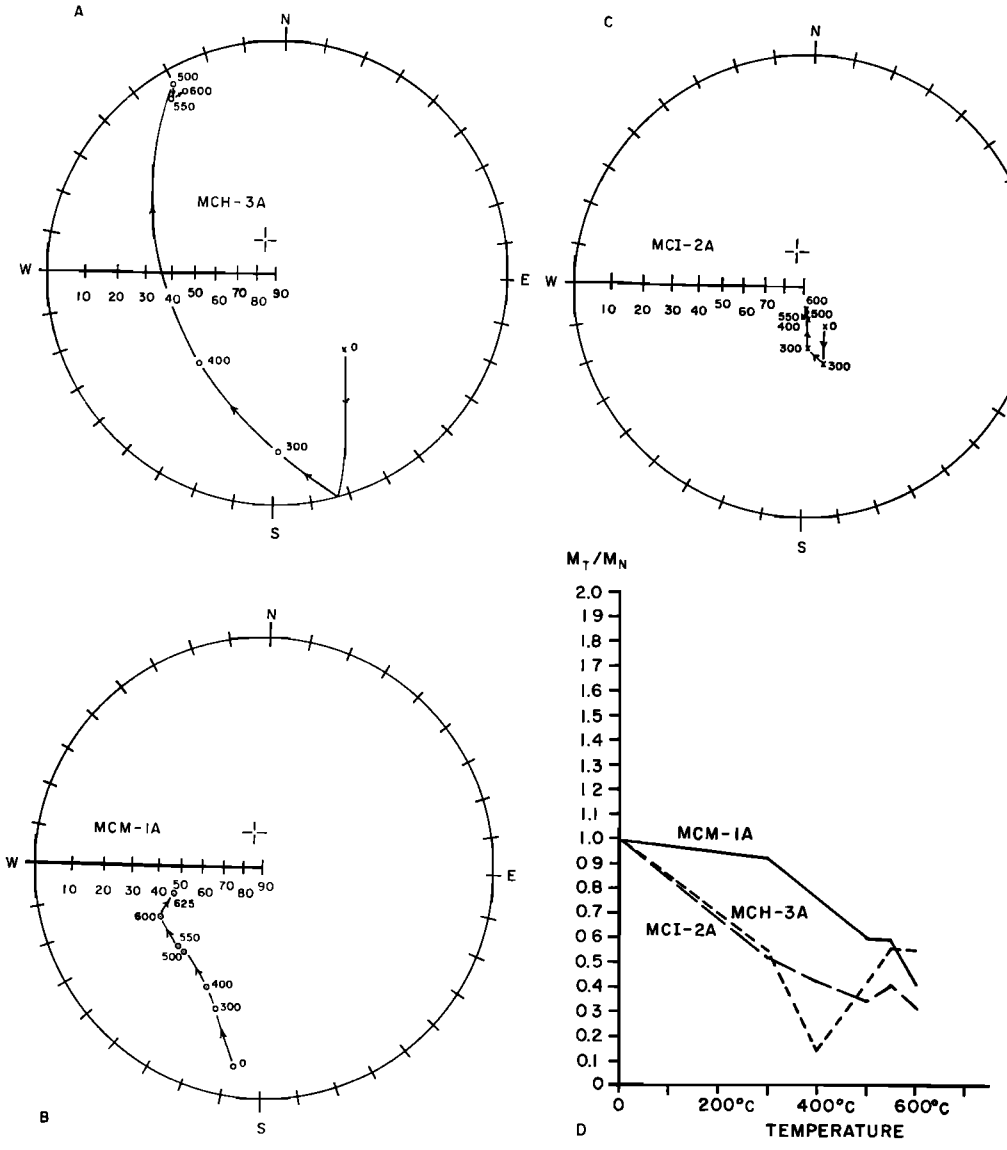


Fig. 4. Examples of thermal demagnetization for samples belonging to groups 2–4, see text. Data corrected for bedding tilt. Present field position indicated by large broken cross.

data after a correction for bedding tilt has been applied. There is a decrease in dispersion after the bedding correction. The Fisher statistics are given in the legend of Figure 6. We can test the decrease in dispersion for significance by the test proposed by *McElhinny* [1964]. To apply the test, the ratio of dispersion before folding to dispersion after folding is computed and compared with values tabulated by *McElhinny* [1964]. For the data from the Mauch Chunk the test is positive ($k_b/k_f = 5.07$) and significant at the 99% confidence level.

These tests indicate that the paleomagnetic direction was acquired before the folding of the Mauch Chunk. These beds were folded during the Appalachian orogeny (Permian). The actual age of magnetization could be anywhere from Upper Mississippian to Lower Permian. Table 2 gives the pole positions for the upper Paleozoic of North America, and Figure 7 is a plot of

21562202, 1968, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB073i020p06515 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

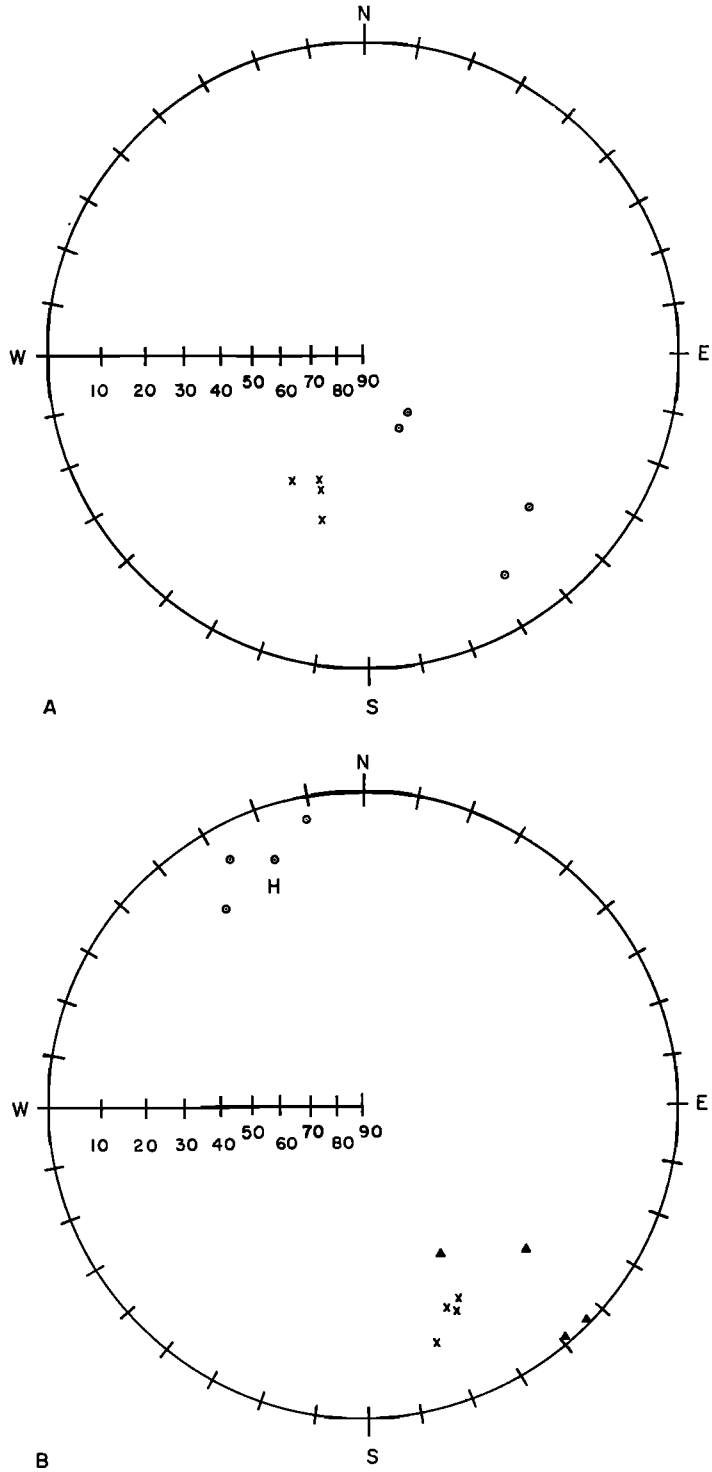


Fig. 5. *Tilt test 1.* (*A*) Thermally cleaned directions from sites MCC (crosses) and MCD (circles, negative inclination) plotted with reference to horizontal. (*B*) Same data as Figure 5*A*, but corrected for bedding tilt. MCC (crosses), MCD (triangles, positive inclination), MCH (circles, negative inclination).

21562202, 1968, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB073i020p06515 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

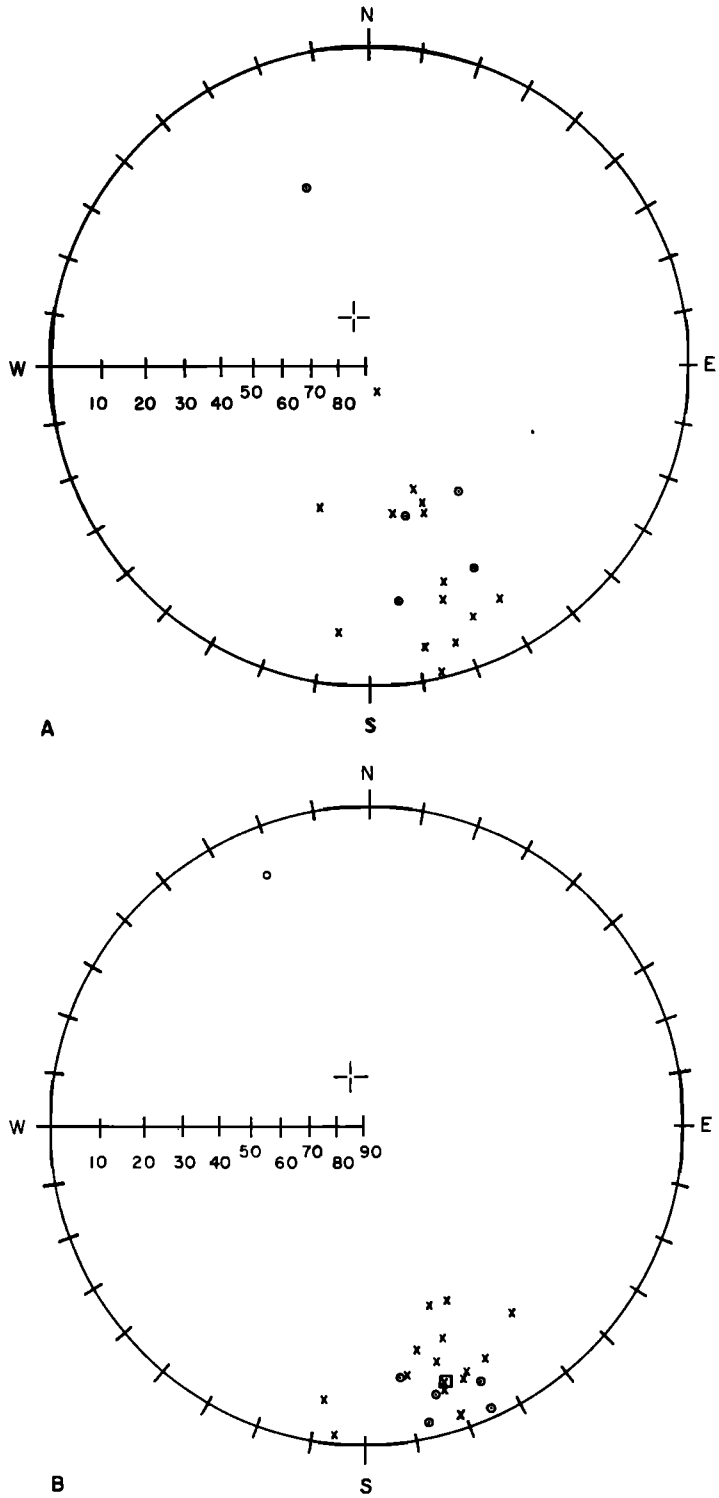


Fig. 6. *Tilt test 2.* (*A*) Site mean directions of thermally cleaned Mauch Chunk referred to horizontal. $D_m = 166°$, $I_m = +4°$, $N = 20$, $R = 16.58$, $K = 5.56$, $\alpha_{95} = 15.23$. (*B*) Same data as Figure 6*A*, but corrected for bedding tilt. $D_m = 162°$, $I_m = +8°$, $N = 20$, $R = 19.32$, $K = 28.21$, and $\alpha_{95} = 6.25$. Mean direction is indicated by box around cross.

21562202, 1968, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB073i020p06515 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Pole Positions for the Upper Paleozoic of North America

Formation and Age	Location	Treatment	D_m, I_m	Pole Position	Reference
Bloomsburg formation Upper Silurian	40°N, 78°W	Thermal demagnetization	001°, −32°	32°N, 102°E	*Roy et al.* [1967]
Barnett shale Middle Mississippian	39°N, 99°W	NRM	142°, +5°	41°N, 135°E	*Martinez and Howell* [1956]
Mauch Chunk formation Upper Mississippian	40°N, 77°W	Thermal demagnetization	163°, +10°	43°N, 127°E	This paper
Hurley Creek formation Upper Pennsylvanian	46°N, 66°W	Thermal demagnetization	171°, +9°	39°N, 125°E	*Roy et al.* [1968]
Maringouin formation Upper Mississippian	45.7°N, 64.5°W	Thermal demagnetization	179°, +21°	34°N, 117°E	*Roy and Robertson* [1968]
Dunkard formation Permian	39 5°N, 81°W	ac demagnetization	163°, +8°	43°N, 123°E	*Helsley* [1965]
Prince Edward Isle Permian	46.5°N, 64°W	Thermal demagnetization	168°, 0°	42°N, 133°E	*Roy* [1966]
Newark group Triassic	40.5°N, 75°W	ac demagnetization	359°, +25°	63°N, 108°E	*Opdyke* [1961]

these data. The pole position of the Mauch Chunk is plotted in the vicinity of other poles for the Permo-Carboniferous of North America.

Statistical Analysis

The mean directions of magnetization were computed by the statistical methods developed by *Fisher* [1953]. Table 1 gives the mean directions of magnetization according to site, treatment, and geographical location. The Fisher statistics for NRM direction are $D_m = 162°$, $I_m = +10°$, $N = 20$, $R = 19.11$, $K = 21.48$, and $\alpha_{95} = 7.20$, and for the same sites after cleaning they are $D_m = 162°$, $I_m = +8°$, $N = 20$, $R = 19.32$, $K = 28.21$, and $\alpha_{95} = 6.25$. The similarity between the two sets of data suggests that we have removed secondary components that are either subparallel to the primary direction or are of relatively low intensity if oriented in a direction other than the primary direction. The mean direction after cleaning for all sites from the Broadtop syncline is $D_m = 162°$, $I_m = +10°$, and for the sites from the Anthracite region it is $D_m = 163°$, $I_m = +6°$.

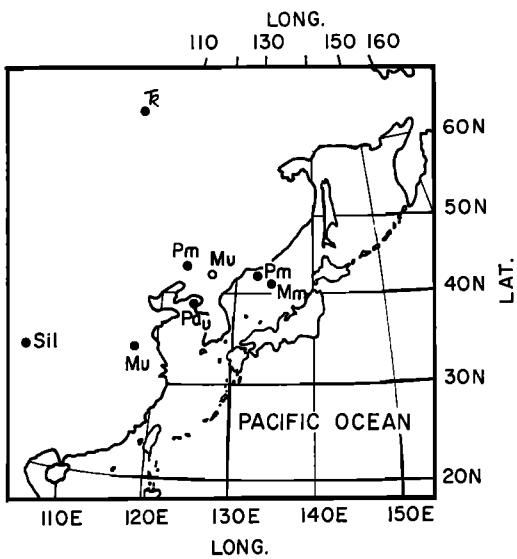


Fig. 7. Pole positions for the upper Paleozoic of North America. Source of data is shown in Table 2.

Tectonic Interpretation of the Data

Figure 8*A* shows the site mean directions of magnetization for thermally cleaned data from the Broadtop syncline, and Figure 8*B* shows the site mean directions of magnetization for thermally cleaned data from the Anthracite region. There is no difference in the paleomagnetic directions from these two localities. Therefore, there has been no relative rotation of these two localities at least since upper Paleozoic time. These data suggest that the hypothesis that the folded Appalachian salient in Pennsylvania was produced by horizontal rotation between these two localities should be rejected. The question of the origin of the salient remains. The paleomagnetic data are consistent with at least two interpretations. The first interpretation (Figure 9) assumes an originally straight geosyncline. The rocks were folded, and then differential transport westward along a series of strike-slip faults produced the salient. This interpretation would require the presence of a number of

21562202, 1968, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB073i020p06515 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

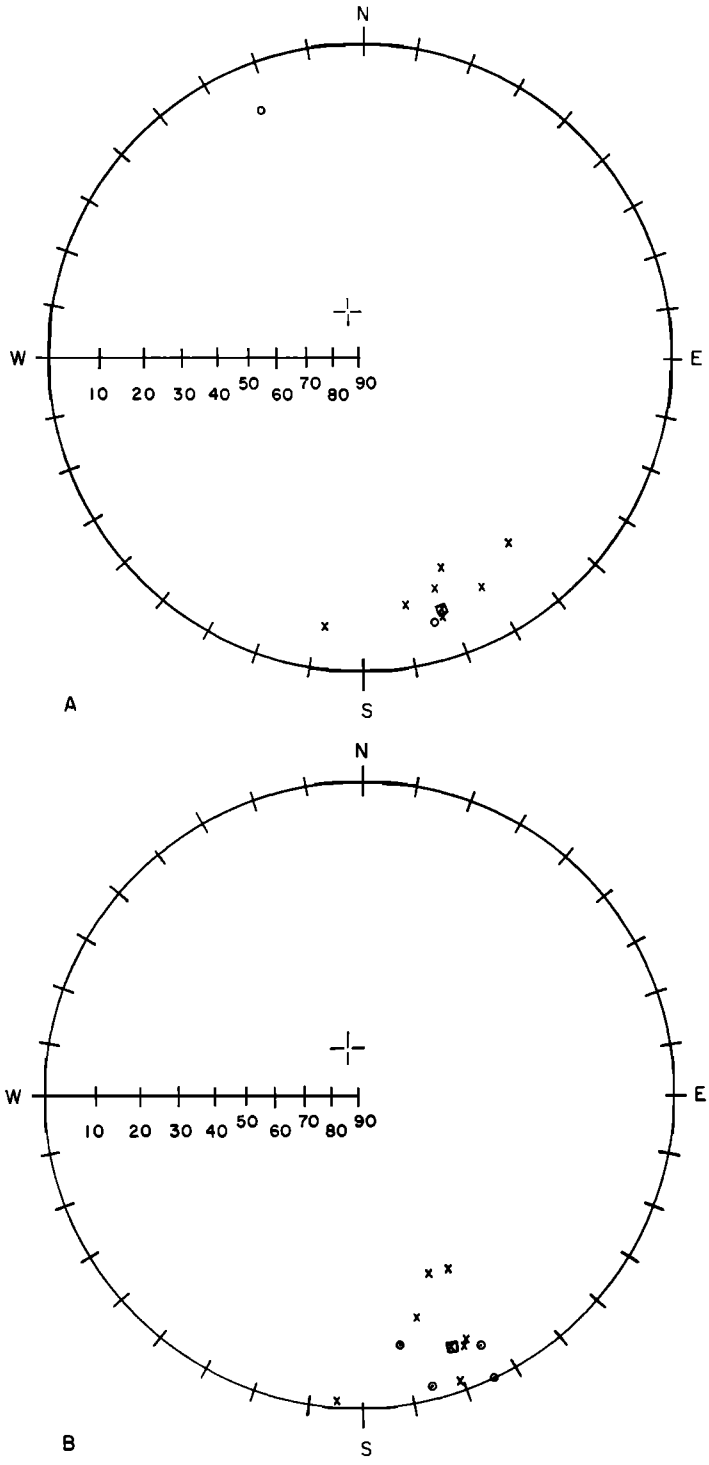


Fig. 8. (*A*) Site mean direction of thermally cleaned data from the Broadtop syncline corrected for bedding tilt. Mean direction is indicated by box around cross. $D_m = 162°$, $I_m = +10°$, $N = 9$, $R = 8.73$, $K = 29.95$. Present field position is indicated by large broken cross. (*B*) Site mean directions of thermally cleaned data from the Anthracite region corrected for bedding tilt. $D_m = 163°$, $I_m = +6°$, $N = 11$, $R = 10.60$, $K = 25.59$.

21562202, 1968, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB073i020p06515 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

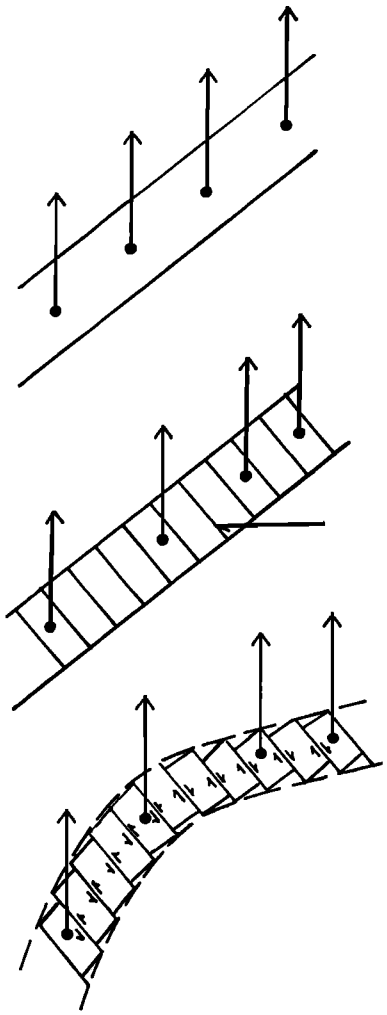


Fig. 9. Formation of curvature in a folded mountain system by differential movement along strike-slip faults. (*Top*) Straight folded geosyncline. Arrows represent hypothetical paleomagnetic direction vectors. (*Middle*) Formation of strike-slip faults. (*Bottom*) Differential displacement and formation of salient. See text.

strike-slip faults in the folded Appalachians. Such faulting does not appear on the geologic map of Pennsylvania (1960 edition). The second interpretation would assume that the present curvature is essentially that of the original geosyncline. The rocks were folded but no bending in plan has taken place.

Roy et al. [1967] state that, although their data from the Silurian Bloomsburg formation are not conclusive, they could be interpreted as indicating that the curvature in the folded Appalachians follows closely that of the original geosyncline. The paleomagnetic data from the Mauch Chunk appear to support a similar interpretation.

Acknowledgments. We would like to thank Professors Ian W. D. Dalziel and John E. Nafe for critically reading the manuscript. Miss Kathryn Wilson wrote and executed the computer programs used in the study.

This work was supported by the National Science Foundation under grant GP-4004.

References

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc. London, A, 217,* 295–305, 1953.

Foster, John, A paleomagnetic spinner magnetometer using a fluxgate gradiometer, *Earth Planetary Sci. Letters, 1,* 463–466, 1966.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, *J. Geophys. Res., 54,* 131–167, 1949.

Gwinn, Vinton, Curvature of marginal folded belts flanking major mountain ranges: Accented or caused by lateral translation of epidermal stratified cover? (abstract), *Program Geol. Soc. Am.,* 87, 1967.

Helsley, C. E., Paleomagnetic results from the Lower Permian Dunkard series of West Virginia, *J. Geophys. Res., 70,* 413–424, 1965.

Irving, E., *Paleomagnetism,* p. 399, John Wiley, New York, 1964.

Irving, E., and N. D. Opdyke, The paleomagnetism of the Bloomsburg red beds, and its possible application to the tectonic history of the Appalachians, *Geophys. J., 9,* 153–167, 1965.

McElhinny, M. W., Statistical significance of the fold test in paleomagnetics, *Geophys. J., 8,* 338–340, 1964.

Martinez, J. D., and L. G. Howell, Paleomagnetism of chemical sediments, *Nature, 178,* 204–205, 1956.

Opdyke, N. D., The paleomagnetism of the New Jersey Triassic: A field study of the inclination error in red sediments, *J. Geophys. Res., 66,* 1941–1949, 1961.

Roy, J. L., Désaimantation thermique et analyse statistique des directions de sédiments carbonifères et permiens de l'est du Canada, *Can. J. Earth Sci., 3,* 139–161, 1966.

Roy, J. L., N. D. Opdyke, and E. Irving, Further paleomagnetic results from the Bloomsburg formation, *J. Geophys. Res., 72,* 5075–5086, 1967.

Roy, J. L., and W. A. Robertson, Evidence of a diagenetic remanent magnetization in the Ma-

21562202, 1968, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB073i020p06515 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

ringouin formation, *Can. J. Earth Sci., 5,* 275–285, 1968.

Roy, J. L., W. A. Robertson, and J. K. Park, Stability of the magnetization of the Hurley Creek formation, *J. Geophys. Res., 73,* 697–702, 1968.

Thellier, E., Sur l'aimantation des terres cuites et ses applications géophysique, Thesis, University of Paris, 1938.

(Received April 4, 1968.)

21562202, 1968, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB073i020p06515 by Curtin University Library, Wiley Online Library on [26/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License